

Laplace Approximation & a Taste of Sampling

INFO 521 · student notes · Module 5 · Lecture B · ~7 min read

Lecture A left a posterior we can evaluate anywhere but cannot normalize. Two escape routes recover approximate inference: replace it with a Gaussian, or draw from it. The running example is deliberately small — 25 real NHANES adults — because at large N every method agrees and there is nothing to see.

Laplace: a Gaussian at the peak

Find the peak (the MAP, by Newton), Taylor-expand the log-posterior to second order there, and declare the quadratic exact. A quadratic log-density is a Gaussian, so

$$p(\mathbf{w} \mid \mathcal{D}) \approx \mathcal{N}(\hat{\mathbf{w}}_{\text{MAP}}, \mathbf{A}^{-1}), \quad \mathbf{A} = \mathbf{X}^\top \mathbf{R} \mathbf{X} + \frac{1}{\tau^2} \mathbf{I}.$$

Two computable ingredients — a Newton fit and one matrix inverse — and all Bayesian machinery downstream reuses the Module 4 Gaussian formulas.

The through-line closes here. The curvature matrix $\mathbf{X}^\top \mathbf{R} \mathbf{X}$ is the observed Fisher information, and $\mathbf{X}^\top \mathbf{X}$ has now appeared four times: Module 1 (invertible $\Rightarrow$ unique $\hat{\mathbf{w}}$), Module 3 (the Hessian, and $\mathcal{I}^{-1} = \text{Cov}[\hat{\mathbf{w}}]$), Module 4 (the data's precision in $\mathbf{S}_N^{-1}$), and Module 5 (the Laplace covariance $^{-1}$). One object — *curvature: how sharply the data pin the parameters down* — the whole course through.

Laplace is **local**: it matches the peak and its curvature and nothing else. On 25 patients the exact posterior for the age slope is skewed with a heavy upper tail, and the symmetric Gaussian misprices exactly there. It fails on skew and multimodality, but **sharpens with data** (posteriors Gaussian-ize as N grows) — cheap and excellent at scale, risky at $N = 25$.

Metropolis: draw instead of describe

If we had samples $\mathbf{w}^{(s)} \sim p(\mathbf{w} \mid \mathcal{D})$, every integral would become an average — intervals from sample quantiles, predictions by averaging $\sigma(\mathbf{w}^{(s)\top} \mathbf{x}_*)$, no normalizer ever computed. **Random-walk Metropolis** draws them: propose a nearby move, and accept it with probability equal to the ratio of the (unnormalized) posterior at the two points. Because it is a ratio, the intractable normalizer **cancels**. Uphill moves are always taken, downhill moves sometimes — that “sometimes” is what makes it *explore* rather than *optimize*.

On the 25-patient cohort the trace is a stationary “hairy caterpillar” (good mixing), and the histogram of draws reproduces the exact skewed posterior — including the tail Laplace flattened — for free.

Choosing

Laplace costs one Newton run and outputs a Gaussian; its error shrinks with **data**, and it fails on skew/multimodality. Metropolis costs thousands of evaluations and outputs draws; its error shrinks with **compute**, and it fails on poor tuning/slow mixing. Practically: **Laplace first**, sample when the

answer matters in the tails — the two agreeing is itself evidence. (Three sampling knobs to know: burn-in, step size, and correlation between draws.)

What to remember

- Laplace = $\mathcal{N}(\hat{\mathbf{w}}_{\text{MAP}}, \mathbf{A}^{-1})$ with $\mathbf{A} = \mathbf{X}^\top \mathbf{R} \mathbf{X} + \tau^{-2} \mathbf{I}$.
- It matches the peak but misses skew and multimodality; it improves with more data.
- Metropolis accepts on *ratios*, so the normalizer cancels; draws replace formulas.
- Assumptions (Laplace) vs compute (MCMC) is the trade.

Through-lines

This closes the inference arc — loss $\rightarrow$ likelihood $\rightarrow$ posterior $\rightarrow$ approximation — around one recurring matrix. Modules 6–7 change the *question* (drop the labels), not the machinery.